SQL Constraints Assignment

Task 1

Problem Statement

Create a table named Products with the following constraints:

• product_id – Primary Key

• product_name – Cannot be NULL

• price – Must be greater than 0 (CHECK constraint)

• category – Default value should be 'accessories'

Solution

Create Products Table

```
CREATE TABLE Products (
product_id INT PRIMARY KEY,
product_name VARCHAR(100) NOT NULL,
price DECIMAL(10,2) CHECK (price > 0),
category VARCHAR(50) DEFAULT 'accessories'
);
```

Insert Sample Records

```
INSERT INTO Products VALUES
(1, 'Mouse', 500, 'electronics'),
(2, 'Keyboard', 800, 'electronics'),
(3, 'USB Cable', 150, DEFAULT),
(4, 'Headphones', 1500, 'audio'),
(5, 'Laptop Stand', 1200, DEFAULT);
```

Task 2

Problem Statement

Create two tables Department and Employee with a relationship between them.

Department Table

• dept_id – Primary Key

• dept_name

Employee Table

• emp_id – Primary Key

• name

• dept_id – Foreign Key referencing Department table

Solution

Create Department Table

```
CREATE TABLE Department (
dept_id INT PRIMARY KEY,
dept_name VARCHAR(100)
);
```

Create Employee Table

```
CREATE TABLE Employee (
emp_id INT PRIMARY KEY,
name VARCHAR(100),
dept_id INT,
FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
);
```

Task 3

Problem Statement

Insert 5 records into the tables and demonstrate how different constraints generate errors when invalid data is inserted.

Solution

Insert Records into Department

```
INSERT INTO Department VALUES
(1, 'HR'),
(2, 'Finance'),
(3, 'IT'),
(4, 'Marketing'),
(5, 'Sales');
```

Insert Records into Employee

```
INSERT INTO Employee VALUES
(101, 'Rahul', 1),
(102, 'Anita', 2),
(103, 'Kiran', 3),
(104, 'Priya', 4),
(105, 'Arjun', 5);
```

Testing Constraint Errors

1. Primary Key Constraint Error

Trying to insert duplicate product_id

```
INSERT INTO Products VALUES
(1, 'Monitor', 5000, 'electronics');
```

Result:

Error occurs because product_id must be unique.

2. NOT NULL Constraint Error

```
INSERT INTO Products(product_id, price)
VALUES (6, 1000);
```

Result:

Error occurs because product_name cannot be NULL.

3. CHECK Constraint Error

INSERT INTO Products VALUES

(7, 'Tablet', -500, 'electronics');

Result:

Error occurs because price must be greater than 0.

4. DEFAULT Constraint Example

INSERT INTO Products(product_id, product_name, price)

VALUES (8, 'Charger', 700);

Result:

Category will automatically be stored as 'accessories'.

5. Foreign Key Constraint Error

Trying to insert employee with a department that does not exist.

INSERT INTO Employee VALUES

(106, 'Vikas', 10);

Result:

Error occurs because dept_id 10 does not exist in Department table.

SQL Joins and Aggregation Assignment

Problem Statement

Create two tables Employee and Departments with the following structure:

Employee Table

• emp_id

• name

• dept_id

• salary

• address

Departments Table

• dept_id

• dept_name

Insert sample records into both tables and write SQL queries to perform the following operations:

1. Find the employee name and department name

2. Find employees who do not belong to any department

3. Find departments without employees

4. Find employee count per department

5. Find the highest salary in each department

6. Find departments with more than 2 employees

7. Find employees working in the IT department

8. Find employees working in the HR department

9. Find employees working in the Finance department

10. Find employees belonging to the Marketing department

11. Find employees belonging to non-IT departments

Solution

Step 1: Create Tables

```
CREATE TABLE Departments (
dept_id INT PRIMARY KEY,
dept_name VARCHAR(50)
);
CREATE TABLE Employee (
```

```
emp_id INT PRIMARY KEY,
name VARCHAR(50),
dept_id INT,
salary INT,
address VARCHAR(100),
FOREIGN KEY (dept_id) REFERENCES Departments(dept_id)
);
```

Step 2: Insert Sample Data

Insert Departments

```
INSERT INTO Departments VALUES
(1,'IT'),
(2,'HR'),
(3,'Finance'),
(4,'Marketing'),
(5,'Sales');
```

Insert Employees

```
INSERT INTO Employee VALUES
(101,'Rahul',1,60000,'Bangalore'),
(102,'Anita',2,50000,'Mysore'),
(103,'Kiran',1,70000,'Delhi'),
(104,'Priya',3,55000,'Mumbai'),
(105,'Arjun',4,48000,'Chennai'),
(106,'Sneha',1,65000,'Hyderabad'),
(107,'Vikas',NULL,45000,'Pune');
```

Queries

1. Find Employee Name and Department Name

```
SELECT e.name, d.dept_name
FROM Employee e
INNER JOIN Departments d
ON e.dept_id = d.dept_id;
```

2. Find Employees Who Do Not Belong to Any Department

```
SELECT name
FROM Employee
WHERE dept_id IS NULL;
```

3. Find Departments Without Employees

```
SELECT d.dept_name
FROM Departments d
LEFT JOIN Employee e
ON d.dept_id = e.dept_id
WHERE e.emp_id IS NULL;
```

4. Find Employee Count per Department

```
SELECT dept_id, COUNT(emp_id) AS employee_count
FROM Employee
GROUP BY dept_id;
```

5. Find Highest Salary in Each Department

```
SELECT dept_id, MAX(salary) AS highest_salary
FROM Employee
GROUP BY dept_id;
```

6. Find Departments with More Than 2 Employees

```
SELECT dept_id, COUNT(emp_id) AS employee_count
FROM Employee
GROUP BY dept_id
HAVING COUNT(emp_id) > 2;
```

7. Employees Working in IT Department

```
SELECT e.name
FROM Employee e
JOIN Departments d
ON e.dept_id = d.dept_id
WHERE d.dept_name = 'IT';
```

8. Employees Working in HR Department

```
SELECT e.name
FROM Employee e
JOIN Departments d
ON e.dept_id = d.dept_id
WHERE d.dept_name = 'HR';
```

9. Employees Working in Finance Department

```
SELECT e.name
FROM Employee e
JOIN Departments d
ON e.dept_id = d.dept_id
WHERE d.dept_name = 'Finance';
```

10. Employees Belonging to Marketing Department

```
SELECT e.name
FROM Employee e
```

```
JOIN Departments d

ON e.dept_id = d.dept_id

WHERE d.dept_name = 'Marketing';
```

11. Employees Belonging to Non-IT Departments

```
SELECT e.name

FROM Employee e

JOIN Departments d

ON e.dept_id = d.dept_id

WHERE d.dept_name != 'IT';
```